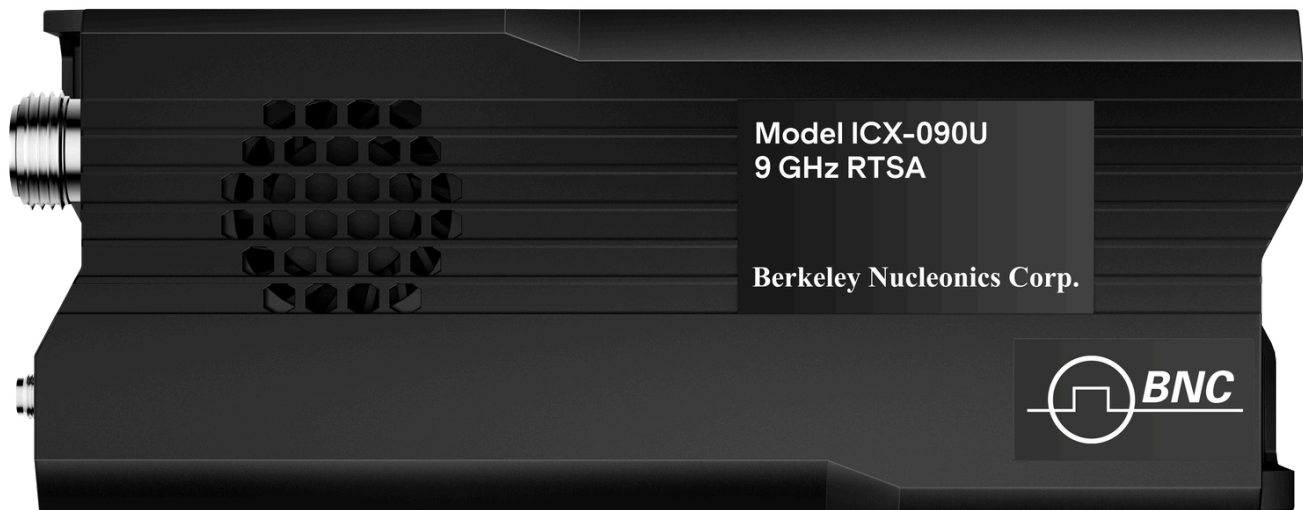


## ICX FIELDHAWK SERIES

# ICX FieldHawk

USB Real-Time Spectrum Analyzers · 9 kHz to 40 GHz



|                                    |                                    |                                      |                                      |                                    |                                   |                                   |
|------------------------------------|------------------------------------|--------------------------------------|--------------------------------------|------------------------------------|-----------------------------------|-----------------------------------|
| <b>ICX-045U</b><br>9 kHz – 4.5 GHz | <b>ICX-060U</b><br>9 kHz – 6.3 GHz | <b>ICX-060U-P</b><br>9 kHz – 6.3 GHz | <b>ICX-080U-P</b><br>9 kHz – 8.5 GHz | <b>ICX-090U</b><br>9 kHz – 9.5 GHz | <b>ICX-200U</b><br>9 kHz – 20 GHz | <b>ICX-400U</b><br>9 kHz – 40 GHz |
|------------------------------------|------------------------------------|--------------------------------------|--------------------------------------|------------------------------------|-----------------------------------|-----------------------------------|

## Features

- Compact USB 3.0/2.0 real-time spectrum analyzer, 9 kHz to 4.5/6.3/8.5/9.5/20/40 GHz
- FPGA-based gapless FFT engine for true real-time spectrum monitoring
- Seven operating modes: standard spectrum, IQ streaming, power detection, real-time spectrum, phase noise, digital demodulation (option), harmonics analysis
- Rich standard measurement set: channel power, OBW, ACPR, IM3, SEM, AM/FM demodulation, automatic phase noise analysis
- Unified API across C/C++, C#, Python, MATLAB, Qt, and LabVIEW on Windows and Linux
- Built-in memory depth of 128 MB for IQ recording and playback

## Applications

- RF testing and interference analysis
- Wireless communications signal monitoring
- RF development and troubleshooting
- Automated test systems (ATE) integration
- Microwave and wideband RF measurements
- Laboratory, aerospace, and defense test

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## OVERVIEW

### Well-Balanced Performance, Cost, and Size

The ICX FieldHawk series features a compact design that delivers excellent RF performance while maintaining cost efficiency. Their outstanding miniaturization allows easy integration into automated test systems, preserving spectrum purity while significantly reducing deployment and operational costs — particularly suited to space- and cost-constrained applications.

### USB Connectivity

ICX FieldHawk analyzers connect over USB, supporting both USB 3.0 and USB 2.0 interfaces for flexible integration with lab PCs, laptops, and embedded controllers.

### Seven Models, Three Tiers

The family spans Economy, Performance+, and Performance++ tiers across six frequency points, with a USB model and an Ethernet model (ICX-FieldHawk-N) at each position. The diagram below maps every model in the family to its interface, performance tier, and maximum frequency; this datasheet's detailed engineering specifications (Spectrum Purity through Input and Output) cover the USB models only.

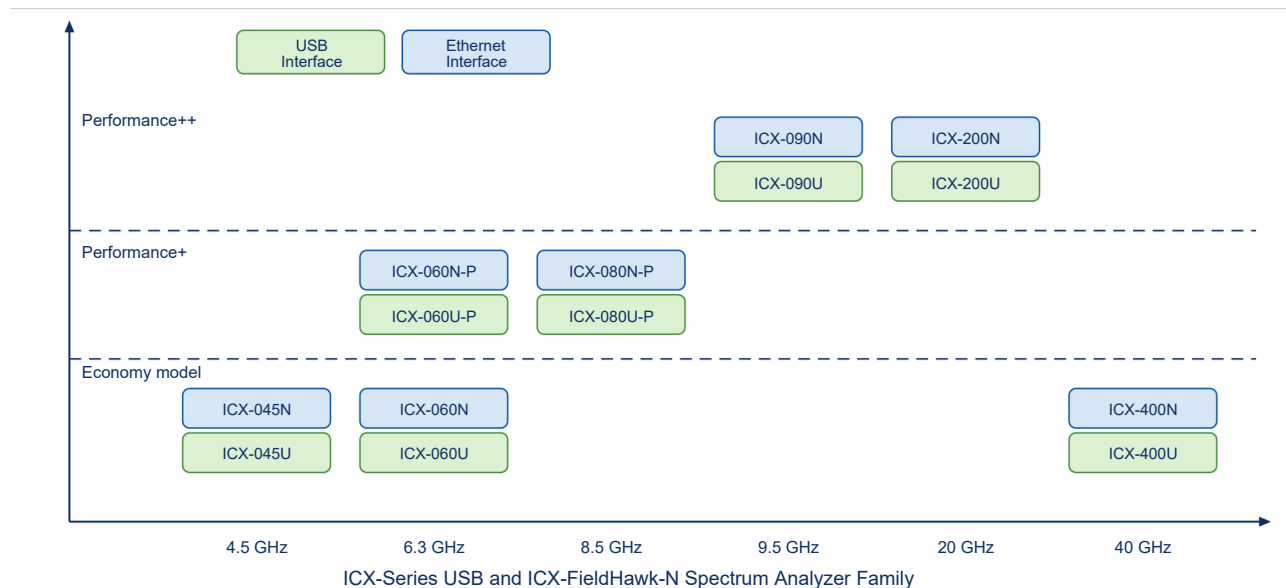


Figure 2-1. Model family overview. Green blocks are USB models (ICX-Series USB); blue blocks are the Ethernet equivalents (ICX-FieldHawk-N).

| Tier          | Frequency        | Model      |
|---------------|------------------|------------|
| Economy       | 9 kHz to 4.5 GHz | ICX-045U   |
| Economy       | 9 kHz to 6.3 GHz | ICX-060U   |
| Performance+  | 9 kHz to 6.3 GHz | ICX-060U-P |
| Performance+  | 9 kHz to 8.5 GHz | ICX-080U-P |
| Performance++ | 9 kHz to 9.5 GHz | ICX-090U   |
| Performance++ | 9 kHz to 20 GHz  | ICX-200U   |
| Economy       | 9 kHz to 40 GHz  | ICX-400U   |

ICX FieldHawk product family overview — see Table of Available Models for descriptions of all seven models. Detailed engineering specifications in this datasheet (Spectrum Purity through Input and Output) cover ICX-045U, ICX-060U, ICX-060U-P, ICX-080U-P, and ICX-400U. ICX-090U and ICX-200U (Performance++ tier) appear in the family table above; their full engineering specifications are tracked separately.

## Unified API

All ICX FieldHawk models share a consistent API, enabling seamless hardware migration without code modification. Development environments include C/C++, C#, Python, MATLAB, Qt, and LabVIEW, compatible with both Windows and Linux.

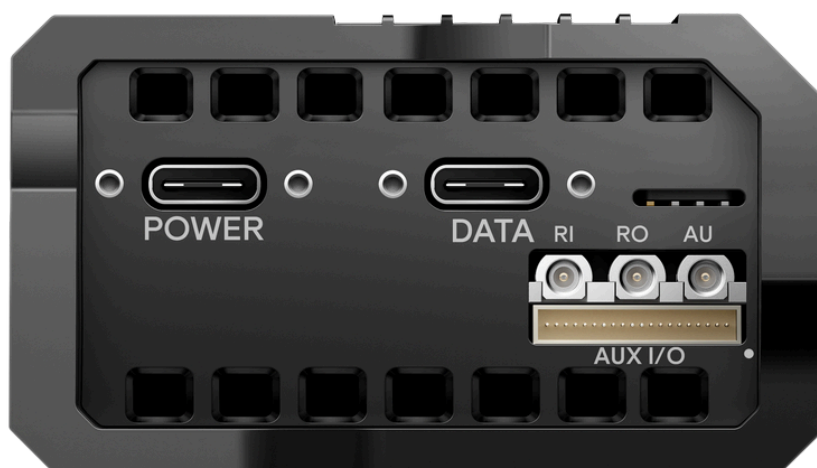
## Rich Standard Measurement Functions

A rich set of advanced measurement capabilities is standard, including channel power, occupied bandwidth, X dB bandwidth, harmonic measurement, SEM, AM/FM demodulation, and automatic phase noise analysis.

# Front Panel



# Back Panel



## MAIN OPERATING MODES

The ICX FieldHawk control software offers seven main operating modes, described below and on the following page.



### Standard Spectrum Analysis Mode

A wide range of measurement functions, including full-span spectrum sweep, channel power, OBW, ACPR, IM3, and SEM. Supports spectrum recording and playback. Combined with auxiliary tools such as signal tracking, peak table, and amplitude correction, it delivers a one-stop platform for comprehensive spectrum checks.



### IQ Streaming Mode

Supports up to 100 MHz analysis bandwidth (model-dependent; see Standard Spectrum Analysis specifications) and allows IQ data acquisition through multiple trigger methods. Provides IQ time-domain waveform display, spectrum and spectrogram views, AM/FM demodulation, and digital down-conversion (DDC).



### Power Detection Analysis Mode

Detection and analysis of time-domain signals within the analysis bandwidth, suited to applications focused on in-band power-versus-time relationships, such as pulse signal measurements.



## MAIN OPERATING MODES (CONTINUED)



### Real-Time Spectrum Analysis Mode

Powered by a high-speed FPGA-based FFTengine, featuring strictly gapless and overlap-free FFT, achieving true real-time monitoring across the full bandwidth.



### Digital Demodulation Mode (Option)

Supports 2ASK, 2FSK, 4FSK, GMSK, BPSK, QPSK, 8PSK, 16QAM, 64QAM, 128QAM, and 256QAM signals.



### Harmonics Analysis Mode

Detection and measurement of up to 10 harmonic components, including harmonic peaks, harmonic channel power, and total harmonic distortion.



### Phase Noise Measurement Mode

Supports offset ranges from 1 Hz to 10 MHz for evaluating carrier phase stability. A built-in automatic carrier search function quickly locates the target carrier without manual adjustment.

# MAIN FUNCTIONS OVERVIEW



MAIN FUNCTIONS OVERVIEW (CONTINUED)





## MAIN FUNCTIONS OVERVIEW (CONTINUED)



Antenna Factor



Amplitude Offset



Signal Track



Peak Table



Data Record and Playback



Multiple Unit Display

## SPECIFICATIONS

All specifications carried over from the OEM source manual (Ed. V1.8, 2026/03/13). Pending BNC engineering verification before external publication. Conditions: 10-minute warm-up, 25°C ambient (50°C core), standard spectrum analysis mode with spurious rejection standard on, adequate heat dissipation maintained.

### Frequency

| Model      | Frequency Range  |
|------------|------------------|
| ICX-045U   | 9 kHz to 4.5 GHz |
| ICX-060U   | 9 kHz to 6.3 GHz |
| ICX-060U-P | 9 kHz to 6.3 GHz |
| ICX-080U-P | 9 kHz to 8.5 GHz |
| ICX-090U   | 9 kHz to 9.5 GHz |
| ICX-200U   | 9 kHz to 20 GHz  |
| ICX-400U   | 9 kHz to 40 GHz  |

Detailed performance specifications below (Spectrum Purity through Input and Output) cover ICX-045U, ICX-060U, ICX-060U-P, ICX-080U-P, and ICX-400U. ICX-090U and ICX-200U (Performance++ tier) are listed in Table of Available Models; their engineering specifications are tracked separately.

|                                        |                                     |
|----------------------------------------|-------------------------------------|
| Reference clock                        | Internal or external                |
| <b>Frequency Accuracy</b>              |                                     |
| TCXO (standard)                        | <1 ppm; manual correction available |
| OCXO (Option 01)                       | <1 ppm; manual correction available |
| GNSS-disciplined OCXO (Options 23/06)  | <0.05 ppm when locked to GNSS       |
| <b>Aging and Temperature Stability</b> |                                     |
| TCXO (standard)                        | <1 ppm/year, <1 ppm                 |
| OCXO (Option 01)                       | <1 ppm/year, <0.15 ppm              |
| GNSS-disciplined OCXO (Options 23/06)  | <1 ppm/year, <0.05 ppm              |

## SPECTRUM PURITY

### Residual Response (dBm)

RBW=1kHz, PosPeakdetector. R.L.=Reference Level.

| Frequency range    | ICX-045U<br>(R.L. 0 / -50 dBm) | ICX-060U<br>(R.L. 0 / -50 dBm) | ICX-060U-P<br>(R.L. 0 / -50 dBm) | ICX-080U-P<br>(R.L. 0 / -50 dBm) | ICX-400U<br>(R.L. 0 / -50 dBm) |
|--------------------|--------------------------------|--------------------------------|----------------------------------|----------------------------------|--------------------------------|
| 100 kHz to 100 MHz | -85 / -110                     | -90 / -110                     | -101 / -123                      | -99 / -122                       | -72 / -103                     |
| 100 MHz to 4.5 GHz | -85 / -110                     | -90 / -110                     | -87 / -116                       | -88 / -119                       | -72 / -103                     |
| 4.5 GHz to 6.3 GHz | —                              | -90 / -110                     | -87 / -116                       | -88 / -119                       | -72 / -103                     |
| 6.3 GHz to 8.5 GHz | —                              | —                              | —                                | -84 / -113                       | -72 / -103                     |
| 8.5 GHz to 10 GHz  | —                              | —                              | —                                | —                                | -72 / -103                     |
| 10 GHz to 20 GHz   | —                              | —                              | —                                | —                                | -91 / -115                     |
| 10 GHz to 40 GHz   | —                              | —                              | —                                | —                                | -85 / -105                     |

Spur reject function: ICX-045U, ICX-060U, ICX-060U-P, and ICX-080U-P — enhanced. ICX-400U — bypass.

### SSB Phase Noise (dBc/Hz)

| Offset  | ICX-045U<br>(1 GHz / 4.5 GHz<br>carrier) | ICX-060U<br>(1 GHz / 6.3 GHz<br>carrier) | ICX-060U-P<br>(1 GHz / 6.3 GHz<br>carrier) | ICX-080U-P<br>(1 GHz / 8.5 GHz<br>carrier) | ICX-400U<br>(1 GHz / 40 GHz<br>carrier) |
|---------|------------------------------------------|------------------------------------------|--------------------------------------------|--------------------------------------------|-----------------------------------------|
| 1 kHz   | -103.4 / -93.5                           | -105.2 / -91.2                           | -107.5 / -92.7                             | -110.3 / -93.5                             | -99.0 / -78.4                           |
| 10 kHz  | -111.3 / -100.3                          | -110.4 / -99.3                           | -114.2 / -99.7                             | -120.0 / -100.5                            | -107.5 / -85.7                          |
| 100 kHz | -109.3 / -98.5                           | -110.5 / -97.4                           | -112.5 / -98.6                             | -120.1 / -100.8                            | -107.7 / -85.1                          |
| 1 MHz   | -129.5 / -121.9                          | -130.1 / -119.9                          | -132.8 / -120.1                            | -131.4 / -116.9                            | -122.7 / -100.8                         |

### Image & IF Rejection, IIP3/IIP2

| Parameter                                       | ICX-045U / ICX-060U / ICX-060U-P /<br>ICX-080U-P                                                                                                                            | ICX-400U                                                                               |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Image rejection, typical<br>(standard / bypass) | 90 MHz to 4.5/6.3/8.5 GHz: >90 dBc<br>/ >35 dBc                                                                                                                             | 8.5–33 GHz: >90 dBc (no bypass rejection); 33–40 GHz:<br>>58 dBc (no bypass rejection) |
| IF rejection, typical                           | Low-IF architecture: 8.2 to 21.75 GHz >68 dBc, other bands >90 dBc. LO-related spurious <-65<br>dBc at center frequency $\pm(N/M) \times 125$ MHz (N, M = 1, 2, 3, 4, 5...) |                                                                                        |

| Model      | Carrier | R.L. = 20 dBm | R.L. = 0 dBm | R.L. = -20 dBm |
|------------|---------|---------------|--------------|----------------|
| ICX-045U   | 1 GHz   | 47.4 / 85.8   | 35.1 / 85.5  | 10.0 / 66.3    |
|            | 4.5 GHz | 45.6 / 98.0   | 26.1 / 91.6  | 6.9 / 19.4     |
| ICX-060U   | 1 GHz   | 46.6 / 86.0   | 29.6 / 85.8  | 10.5 / 67.3    |
|            | 6.3 GHz | 42.9 / 109.5  | 24.6 / 98.5  | 3.9 / 17.1     |
| ICX-060U-P | 1 GHz   | 51.0 / 84.9   | 40.1 / 85.1  | 10.0 / 66.4    |
|            | 6.3 GHz | 43.4 / 65.9   | 25.3 / 94.6  | 4.7 / 17.7     |
| ICX-080U-P | 1 GHz   | 49.6 / 87.5   | 35.6 / 84.3  | 11.5 / 67.4    |
|            | 8.5 GHz | 41.0 / 57.4   | 25.5 / 44.8  | 2.4 / 34.2     |
| ICX-400U   | 1 GHz   | 40.3 / 75.5   | 27.4 / 45.3  | 8.7 / 25.2     |
|            | 40 GHz  | 31.7 / 88.6   | 10.3 / 86.1  | 4.8 / 66.6     |

IIP3/IIP2 values shown as IIP3 / IIP2 (dBm).

## AMPLITUDE

| Parameter              | ICX-045U / ICX-060U / ICX-060U-P / ICX-080U-P                                       | ICX-400U                                                                   |
|------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Display range          | DANL to +23 dBm                                                                     | DANL to +20 dBm                                                            |
| Reference level (R.L.) | -50 dBm to +23 dBm                                                                  | -50 dBm to +20 dBm                                                         |
| Amplitude accuracy     | 9 kHz to 4.5/6.3/8.5 GHz: $\pm 2.0$ dB                                              | 9 kHz to 9.5 GHz: $\pm 2.0$ dB; 9.5 to 40 GHz: $\pm 3.0$ dB                |
| Max. input power (CW)  | 23 dBm, 30 MHz to 4.5/6.3/8.5 GHz, preamp off; 10 dBm, 9 kHz to 30 MHz or preamp on | 23 dBm, 50 MHz to 40 GHz, preamp off; 10 dBm, 9 kHz to 50 MHz or preamp on |
| VSWR                   | 30 MHz to 4.5/6.3/8.5 GHz: $< 2.5:1$                                                | 90 MHz to 16 GHz: $< 2.0:1$ ; 16 to 40 GHz: $< 3.0:1$                      |
| Max. DC voltage        | $\pm 10$ VDC                                                                        |                                                                            |
| IF in-band flatness    | $\pm 2.0$ dB                                                                        |                                                                            |
| RF preamplifiers       | Automatically turn on, or forcibly turn off                                         |                                                                            |

## Display Average Noise Level (DANL, dBm/Hz, RBW = 1 kHz)

| Frequency range    | ICX-045U<br>(R.L. -20 / -50 dBm) | ICX-060U<br>(R.L. -20 / -50 dBm) | ICX-060U-P<br>(R.L. -20 / -50 dBm) | ICX-080U-P<br>(R.L. -20 / -50 dBm) | ICX-400U<br>(R.L. -20 / -50 dBm) |
|--------------------|----------------------------------|----------------------------------|------------------------------------|------------------------------------|----------------------------------|
| 9 kHz to 1 MHz     | -134.6 / -150.3                  | -136.4 / -147.9                  | -135.9 / -148.5                    | -141.4 / -151.7                    | -136.0 / -145.8                  |
| 1 MHz to 30 MHz    | -140.2 / -162.6                  | -139.7 / -162.3                  | -140.7 / -162.8                    | -154.2 / -161.6                    | -153.7 / -158.0                  |
| 30 MHz to 3.0 GHz  | -153.2 / -163.5                  | -152.7 / -164.8                  | -152.1 / -163.9                    | -150.8 / -167.1                    | -153.7 / -158.0                  |
| 3.0 GHz to 4.5 GHz | -155.2 / -162.7                  | -157.1 / -163.5                  | -151.3 / -162.0                    | -155.6 / -164.7                    | -154.1 / -159.9                  |
| 4.5 GHz to 6.3 GHz | —                                | -151.9 / -160.4                  | -151.3 / -162.0                    | -155.6 / -164.7                    | -154.1 / -159.9                  |
| 6.3 GHz to 8.5 GHz | —                                | —                                | —                                  | -144.0 / -157.2                    | -154.1 / -159.9                  |
| 8.5 GHz to 19 GHz  | —                                | —                                | —                                  | —                                  | -154.1 / -159.9                  |
| 19 GHz to 40 GHz   | —                                | —                                | —                                  | —                                  | -145.2 / -149.3                  |



## STANDARD SPECTRUM ANALYSIS

| Parameter    | ICX-045U/ICX-060U,<br>ICX-060U-P/ICX-080U-P                                     | ICX-400U         |
|--------------|---------------------------------------------------------------------------------|------------------|
| RBW          | 0.1 Hz to 2.5 MHz                                                               | 0.1 Hz to 10 MHz |
| VBW          | 0.1 Hz to 10 MHz                                                                |                  |
| Detector     | PosPeak, NegPeak, Sample, Average, RMS, MaxPower                                |                  |
| Data chart   | SpectraCore software provides spectrum, spectrogram, and historical trace views |                  |
| Measurements | Channel power, OBW, X dB bandwidth, adjacent channel power ratio, IM3           |                  |

## Sweep Speed

| Model      | RBW 250 kHz, FPGA, spur reject bypass | RBW 250 kHz, FPGA, spur reject standard | RBW 50 kHz, FPGA, spur reject bypass | RBW 1 kHz, CPU, spur reject bypass |
|------------|---------------------------------------|-----------------------------------------|--------------------------------------|------------------------------------|
| ICX-045U   | 186.0 GHz/s                           | 86.9 GHz/s                              | 79.0 GHz/s                           | 5.7 GHz/s                          |
| ICX-060U   | 444.5 GHz/s                           | 209.4 GHz/s                             | 157.3 GHz/s                          | 6.2 GHz/s                          |
| ICX-060U-P | 793.3 GHz/s                           | 382.7 GHz/s                             | 242.2 GHz/s                          | 5.9 GHz/s                          |
| ICX-080U-P | 822.1 GHz/s                           | 359.2 GHz/s                             | 230.8 GHz/s                          | 5.8 GHz/s                          |
| ICX-400U   | 1.1 THz/s                             | 584.6 GHz/s                             | 215.6 GHz/s                          | 4.3 GHz/s                          |

## IQ Recording

| Model      | Max. IQ sample rate | Decimation factor     | Max. burst recording bandwidth | Max. continuous recording bandwidth |
|------------|---------------------|-----------------------|--------------------------------|-------------------------------------|
| ICX-045U   | 7.8125 MSPS         | $2^n$ , $n = 0$ to 8  | 6.25 MHz                       | 25 MHz                              |
| ICX-060U   | 31.25 MSPS          | $2^n$ , $n = 0$ to 10 | 25 MHz                         | 50 MHz                              |
| ICX-060U-P | 125 MSPS            | $2^n$ , $n = 0$ to 12 | 100 MHz                        | 50 MHz                              |
| ICX-080U-P | 125 MSPS            | $2^n$ , $n = 0$ to 12 | 100 MHz                        | 50 MHz                              |
| ICX-400U   | 125 MSPS            | $2^n$ , $n = 0$ to 12 | 100 MHz                        | 50 MHz                              |

Built-in memory depth: 128 MB, all models. External trigger response: maximum 500 times/sec (all models).

## DETECTION & REAL-TIME SPECTRUM ANALYSIS

### Detection Analysis

| Model      | Lowest time resolution | Max. analysis bandwidth |
|------------|------------------------|-------------------------|
| ICX-045U   | 128 ns                 | 6.25 MHz                |
| ICX-060U   | 32 ns                  | 25 MHz                  |
| ICX-060U-P | 8 ns                   | 100 MHz                 |
| ICX-080U-P | 8 ns                   | 100 MHz                 |
| ICX-400U   | 8 ns                   | 100 MHz                 |

Detector: PosPeak, NegPeak, Sample, Average, RMS, MaxPower (all models).

### Real-Time Spectrum Analysis

FFT engine implemented in FPGA. Frame compression and trace detection supported; no missing samples between FFT frames. FFT frame update rate =  $10^9 \text{ ns} / (N \times D \times \text{lowest time resolution})$ . POI =  $2 \times N \times D \times \text{lowest time resolution}$ , where N is FFT points (2048, 1024, 512, 256, 128, 64, 32) and D is decimation factor (1, 2, 4, 8...).

| Parameter                     | ICX-045U              | ICX-060U              | ICX-060U-P/ICX-080U-P,<br>ICX-400U |
|-------------------------------|-----------------------|-----------------------|------------------------------------|
| POI, N=2048, D=1              | 524.288 $\mu\text{s}$ | 131.072 $\mu\text{s}$ | 32.768 $\mu\text{s}$               |
| POI, N=32, D=1                | 8.192 $\mu\text{s}$   | 2.048 $\mu\text{s}$   | 0.512 $\mu\text{s}$                |
| FFT refresh rate, N=2048, D=1 | 3,814/sec             | 15,258/sec            | 61,035/sec                         |
| FFT refresh rate, N=32, D=1   | 244,140/sec           | 976,562/sec           | 3,906,250/sec                      |
| Max. analysis bandwidth       | 6.25 MHz              | 25 MHz                | 100 MHz                            |

### RBW (Resolution Bandwidth Windows)

| Window           | ICX-045U            | ICX-060U             | ICX-060U-P/ICX-080U-P,<br>ICX-400U |
|------------------|---------------------|----------------------|------------------------------------|
| Flat-top window  | 920 kHz to 3.59 kHz | 3.68 MHz to 3.59 kHz | 14.73 MHz to 3.59 kHz              |
| B-Nuttall window | 488 kHz to 1.90 kHz | 1.95 MHz to 1.90 kHz | 7.81 MHz to 1.90 kHz               |
| Window grades    | 9                   | 11                   | 13                                 |

Window functions: B-Nuttall, Flat-top, LowSideLobe. Amplitude resolution: 0.75 dB. All models.

## GENERAL

### Power

|        |                                                                                              |
|--------|----------------------------------------------------------------------------------------------|
| Supply | USB Type-C, 5 V 2 A supply capacity; voltage range 4.75 to 5.25 V; ripple less than 200 mVpp |
|--------|----------------------------------------------------------------------------------------------|

### Data

|           |                                                                                   |
|-----------|-----------------------------------------------------------------------------------|
| Interface | USB Type-C, USB 3.0 (USB 2.0 bandwidth-limited); requires 5 V, 0.9 A power supply |
|-----------|-----------------------------------------------------------------------------------|

### GNSS

|                                    |                                                                                                                                                   |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| GNSS type                          | External (accessory)                                                                                                                              |
| GNSS 1PPS synchronization accuracy | ±100 ns (Option 21, standard external GNSS); ±75 ns (Option 22, high-precision external GNSS); ±50 ns (Option 23, external GNSS-disciplined OCXO) |

### Packaging & Accessories

|          |                                                   |
|----------|---------------------------------------------------|
| Included | Flash disk ×1, USB 3.0 cable ×2, power adapter ×1 |
|----------|---------------------------------------------------|

### System Requirements

|                   |                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------|
| Operating systems | Windows 11/10/8/7 (x86, x64); Debian 12/11/10 (x64); Ubuntu 24.04/22.04/20.04/18.04 (x64) |
|-------------------|-------------------------------------------------------------------------------------------|

### Operating / Storage Temperature (Ambient)

|                      |                            |
|----------------------|----------------------------|
| T0 class (standard)  | 0 to 50°C / -20 to +70°C   |
| T1 class (Option 40) | -20 to 65°C / -40 to +85°C |
| T2 class (Option 41) | -40 to 65°C / -40 to +85°C |

### Operating Relative Humidity

|                |          |
|----------------|----------|
| Ambient 0–40°C | 5 to 75% |
| Ambient >40°C  | 5 to 45% |

## INPUT AND OUTPUT

| Connector              | ICX-045U / ICX-060U / ICX-060U-P / ICX-080U-P             | ICX-400U                                                                     |
|------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------|
| External trigger input | USB Type-C, 3.3 V CMOS, high impedance                    | Integrated in AUXIO, 3.3 V CMOS, high impedance                              |
| Trigger output         | USB Type-C, 3.3 V CMOS                                    | Integrated in AUXIO, 3.3 V CMOS                                              |
| RF input               | SMA (F), impedance 50 $\Omega$                            | 2.92 mm (F), impedance 50 $\Omega$                                           |
| RF output              | SMA (F), impedance 50 $\Omega$                            | Not available                                                                |
| Reference clock input  | MCX (F), amplitude $\geq 1.5$ Vpp, impedance 330 $\Omega$ | MMCX (F), amplitude $\geq 1.5$ Vpp, impedance 330 $\Omega$                   |
| Reference clock output | Unavailable                                               | Integrated in AUXIO, 3.3 V CMOS, programmable on/off                         |
| IF output              | Unavailable                                               | MMCX (F), max output -25 dBm, impedance 50 $\Omega$ , 307.2 MHz $\pm 50$ MHz |

## Size, Weight, Power Consumption

| Model                                         | Dimensions       | Weight | Power consumption |
|-----------------------------------------------|------------------|--------|-------------------|
| ICX-045U / ICX-060U / ICX-060U-P / ICX-080U-P | 156 × 62 × 22 mm | <305 g | 9 to 12 W         |
| ICX-400U                                      | 139 × 68 × 31 mm | <420 g | 10 to 14 W        |

Specifications apply under: (1) 10-minute start-up/warm-up; (2) 25°C ambient temperature, 50°C core; (3) standard spectrum analysis mode, spurious rejection standard on; (4) adequate heat dissipation to hold ambient and core temperature within rated range.

## TABLE OF AVAILABLE MODELS

| Model      | Frequency Range  | Description                                                                                                                                     |
|------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| ICX-045U   | 9 kHz to 4.5 GHz | Compact USB spectrum analyzer, ideal for RF testing, interference analysis, and wireless communications applications.                           |
| ICX-060U   | 9 kHz to 6.3 GHz | USB spectrum analyzer providing reliable performance for RF development, troubleshooting, and signal monitoring.                                |
| ICX-060U-P | 9 kHz to 6.3 GHz | Performance-tier USB spectrum analyzer offering enhanced spectrum purity and switching speed for advanced RF testing at 6.3 GHz.                |
| ICX-080U-P | 9 kHz to 8.5 GHz | Performance-tier USB spectrum analyzer extending enhanced spectrum purity and switching speed to 8.5 GHz for advanced RF and microwave testing. |
| ICX-090U   | 9 kHz to 9.5 GHz | Wideband USB spectrum analyzer designed for advanced RF, microwave, and wireless analysis.                                                      |
| ICX-200U   | 9 kHz to 20 GHz  | High-performance USB spectrum analyzer suitable for laboratory, aerospace, defense, and advanced RF applications.                               |
| ICX-400U   | 9 kHz to 40 GHz  | Professional USB spectrum analyzer delivering broad frequency coverage for demanding RF and microwave measurements.                             |

Detailed engineering specifications in this datasheet (Spectrum Purity, Amplitude, Standard Spectrum Analysis, IQ Recording, Detection Analysis, Real-Time Spectrum Analysis, RBW, General, Input and Output) apply to ICX-045U, ICX-060U, ICX-060U-P, ICX-080U-P, and ICX-400U. ICX-090U and ICX-200U (Performance++ tier) are listed here with descriptions only; their full engineering specifications are tracked separately.

## OPTIONS

| Code | Description                                                                                             | Type              |
|------|---------------------------------------------------------------------------------------------------------|-------------------|
| 01   | Built-in OCXO reference clock                                                                           | Built-in hardware |
| 02   | Built-in signal generator (ICX-045U, ICX-060U, ICX-060U-P, ICX-080U-P only — not available on ICX-400U) | Built-in hardware |
| 05   | Internal high-precision GNSS                                                                            | Built-in hardware |
| 06   | Built-in GNSS-disciplined reference clock                                                               | Built-in hardware |
| 20   | AUXIO expansion board                                                                                   | Accessory         |
| 21   | External GNSS                                                                                           | Accessory         |
| 22   | External high-precision GNSS                                                                            | Accessory         |
| 23   | External GNSS-disciplined OCXO reference clock                                                          | Accessory         |
| 34   | External omnidirectional antenna, 400 MHz to 8000 MHz, gain <2 dBi                                      | Accessory         |
| 35   | External active directional antenna, 0.5 to 10 GHz, gain <5 dBi (amp off), <25 dBi (amp on)             | Accessory         |
| 40   | T1 temperature class                                                                                    | Built-in hardware |
| 41   | T2 temperature class                                                                                    | Built-in hardware |
| 71   | Basic digital demodulation                                                                              | Software          |
| 72   | Pulse detection                                                                                         | Software          |

### Built-In Signal Generator (Option 02)

Available for ICX-045U, ICX-060U, ICX-060U-P, and ICX-080U-P only; not available on ICX-400U.

|                    |                                |
|--------------------|--------------------------------|
| Frequency range    | 100 kHz to 6.3 GHz, step 10 Hz |
| Power range        | –50 dBm to 0 dBm, step 0.25 dB |
| VSWR               | 30 MHz to 6.3 GHz: <2.0:1      |
| Non-harmonic spurs | <–50 dBc                       |

| Frequency range   | 2nd harmonic | 3rd harmonic and above |
|-------------------|--------------|------------------------|
| 100 kHz to 30 MHz | <–10 dBc     | <–10 dBc               |
| 30 MHz to 1.6 GHz | <–10 dBc     | <–10 dBc               |
| 1.6 GHz to 3 GHz  | <–20 dBc     | <–20 dBc               |
| 3 GHz to 6.3 GHz  | <–20 dBc     | <–20 dBc               |

Leakage to receiver: >90 dBc (100 kHz–30 MHz), >80 dBc (30 MHz–3 GHz), >70 dBc (3–6.3 GHz).